Knee Complex

Introduction :-

→ It is one of the most often injured jt in the human body.

→ It composed of 2 distinct articulation located within a single jt capsule.

1) Tibiofemoral jt
2) Patellofemoral jt

Tibiofemoral jt - Articulation b/w distal femur and the proximal tibia.

Patellofemoral jt - Articulation b/w posterior patella & the femur.

Tibiofemoral jt Structure :-

→ It is a double condyloid jt.

→ Has 3° of freedom

Flexion / Extension → Sagittal plane, Coronal axis

Medial / Lateral rotation → Transverse plane, Longitudinal axis

Abduction / Adduction → Frontal plane, AP axis.

Femur :-

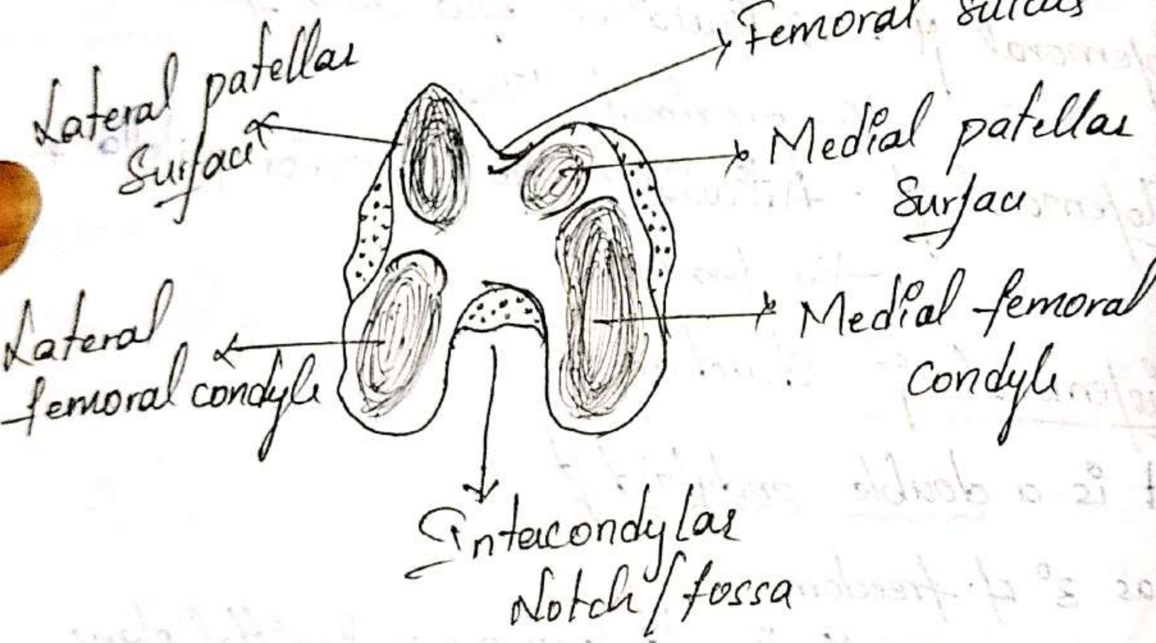
→ Proximally - articulation is of large medial & lateral Condyles of the distal femur.

→ Medial condyle is larger, with a greater radius of Curvature than the lateral condyle.

→ In sagittal plane, condyles have convex surface, with a smaller radius of curvature posteriorly.

→ Tibial articular surface of the lateral femoral condyle is shorter than the tibial articular surface of the medial femoral condyle.

→ Two femoral condyles are seperated inferiorly by Intercondylar notch or fossa.



→ Anteriorly by an asymmetrical, shallow groove is called as Femoral Sulcus, patellar groove, patellar surface.

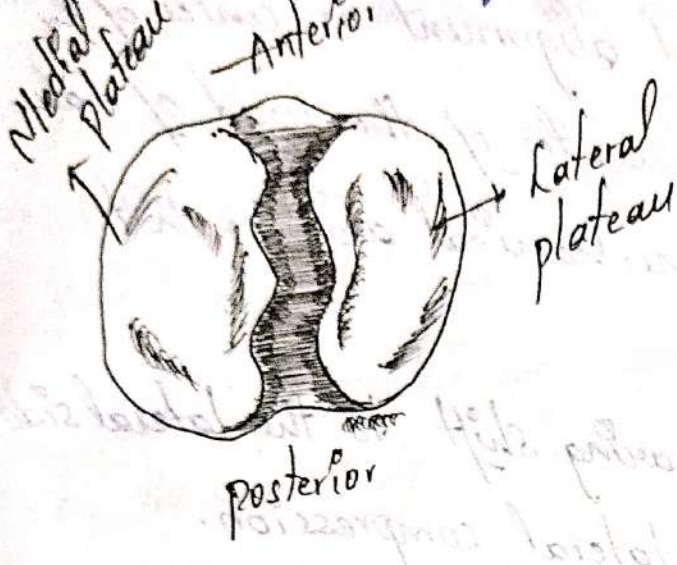
Tibia :-

→ Large convex femoral condyles sit on the flat tibial condyles.

→ Medial tibial plateau is longer in the AP direction than the lateral tibial plateau.

→ Lateral tibial articular cartilage is thicker than the medial side.

→ Tibial plateau slopes posteriorly by 7° to 10°.



→ Medial and lateral tibial condyles are seperated by two bony spines called <u>intercondylar tubercles</u>.

Tibiofemoral Alignment and Wt. bearing forces:-

→ Anatomical or longitudinal axis is directed inferiorly and medially from its proximal & distal end.

→ It is directed vertically.

→ Femoral and tibial longitudinal axis normally an angle medially at the knee is 180° to 185°.

→ Medial tibiofemoral angle ↑s than 185° then the condition is called Genu Valgum or Knock knees.

→ Medial tibiofemoral angle is ↓ then ~~that~~ ~~Condit~~ 180° that condition is known as Genu Varum or bow legs.

→ Measuring tibiofemoral alignment is center of the femoral head to the center of the head of talus

→ This line represents mechanical axis or Wt bearing line.

→ Genu Valgum - wt bearing shift to the lateral side ↑ the lateral compression.

→ Genu Varum - wt bearing shift to the medial side ↑ the medial compression.

Menisci :-

→ It has 2 menisci
1) Lateral menisci
2) Medial Menisci.

→ These menisci plays an imp role in wt. bearing & shock absorbers.

→ Menisci are fibrocartilaginous discs with a semi circular shape.

→ Medial menisci is C-shaped.
Lateral menisci is 4/5th of the ~~cy~~ circle.

→ Both menisci are open toward the intercondylar
... thick peripherally & thin centrally.

Meniscal Attachments :-

→ Anterior & posterior end of the menisci are called the Anterior & posterior horns.

→ Anteriorly, menisci are connected by transverse Ligament.

→ Both menisci are attached to patella through the patellomeniscal ligaments

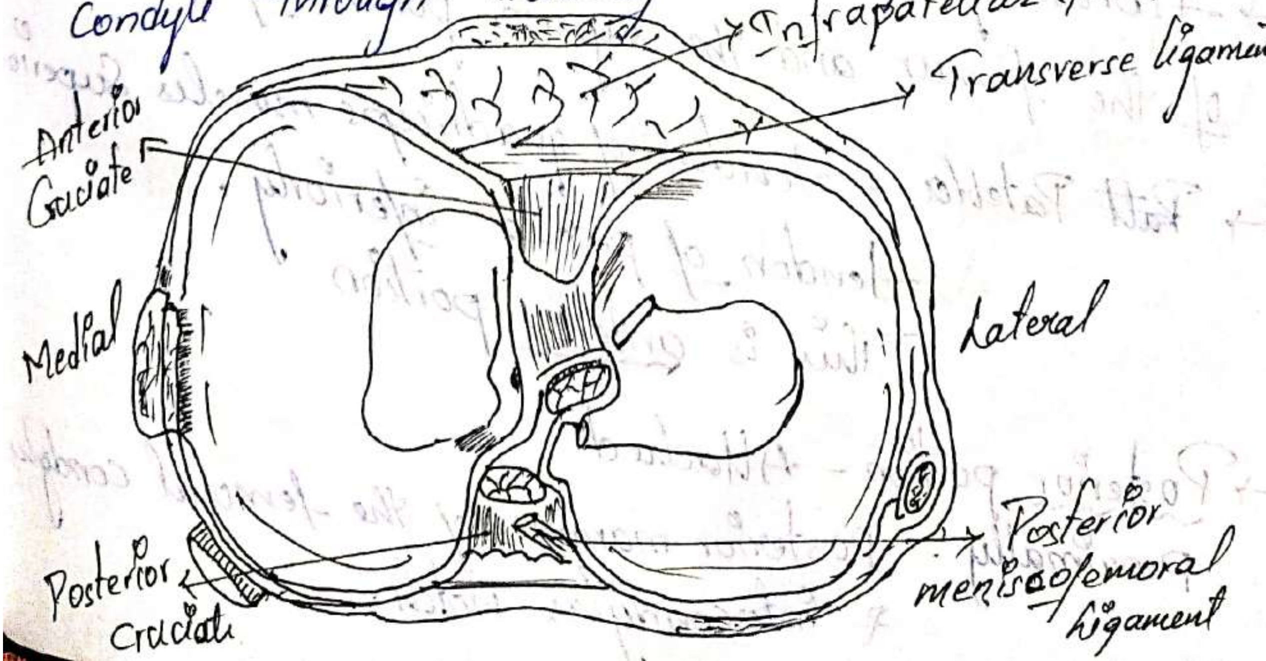
→ Periphery menisci are connected to tibial condyle by the coronal ligaments.

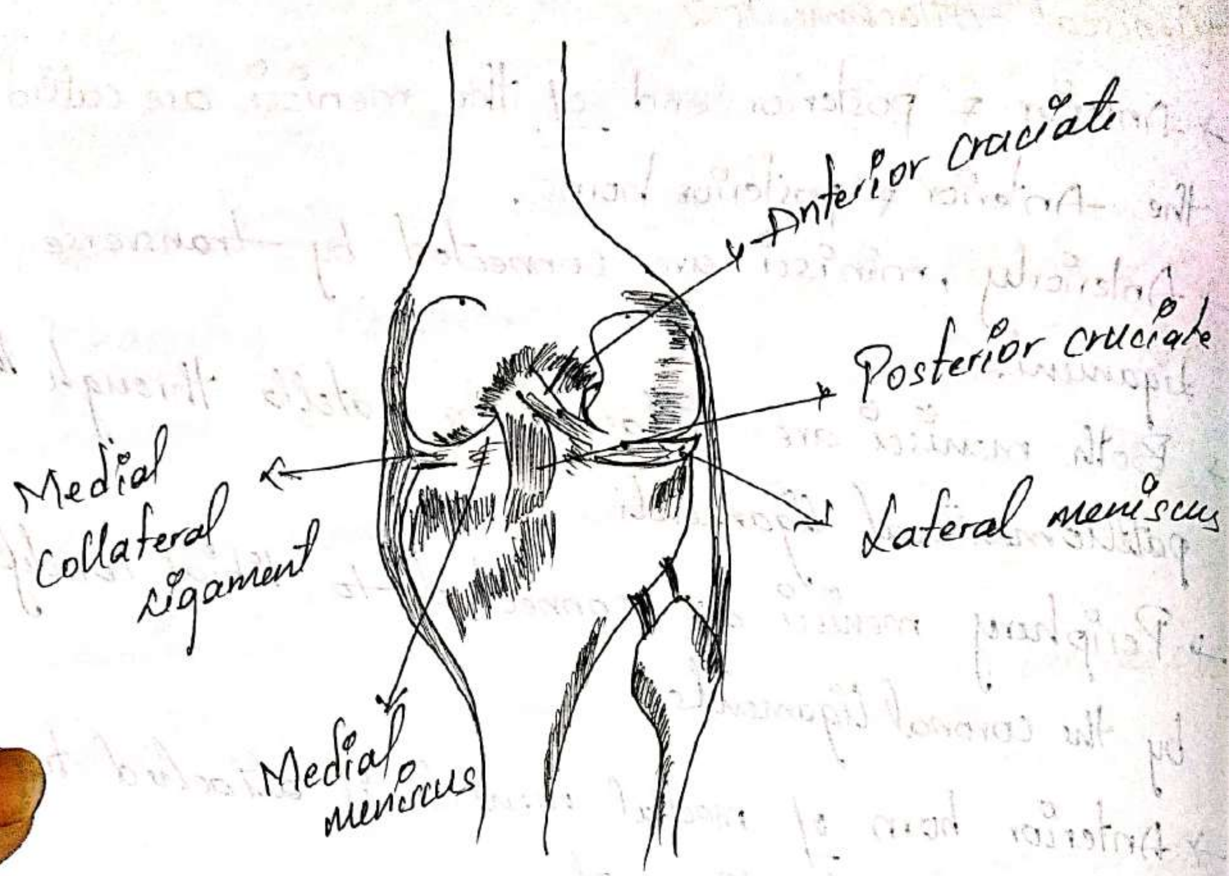
→ Anterior horn of medial menisci is attached to anterior cruciate ligament.

→ Posterior horn of medial menisci is attached to posterior cruciate ligament.

→ Semimembraneous muscle connects to the medial meniscus.

→ Posteriorly, the lateral meniscus attaches to the posterior cruciate ligament and the medial femoral condyle through meniscofemoral ligaments.





Joint Capsule :-

→ Consist of 2 joints.

1) Tibiofemoral jt
2) Patellofemoral jt

→ Superficially - fibrous layer
Deep - Synovial layer or membrane.

→ Fibrous portion - attached to the inferior aspect of the femur and the superior portion of the tibia.

→ ~~Patt~~ Patella - tendon of quadriceps muscles superior
tendon of patella inferiorly.
This is anterior portion

→ Posterior portion - Attached
Proximally - Posterior margine of the femoral condyles
& intercondylar notch.

distally - posterior tibial condyle.

→ Anterior Medial + Anterolateral of the capsule, with medial and lateral patellar retinaculum or Extensor retinaculum.

→ Capsule is reinforced medially, laterally and posteriorly by capsular ligaments

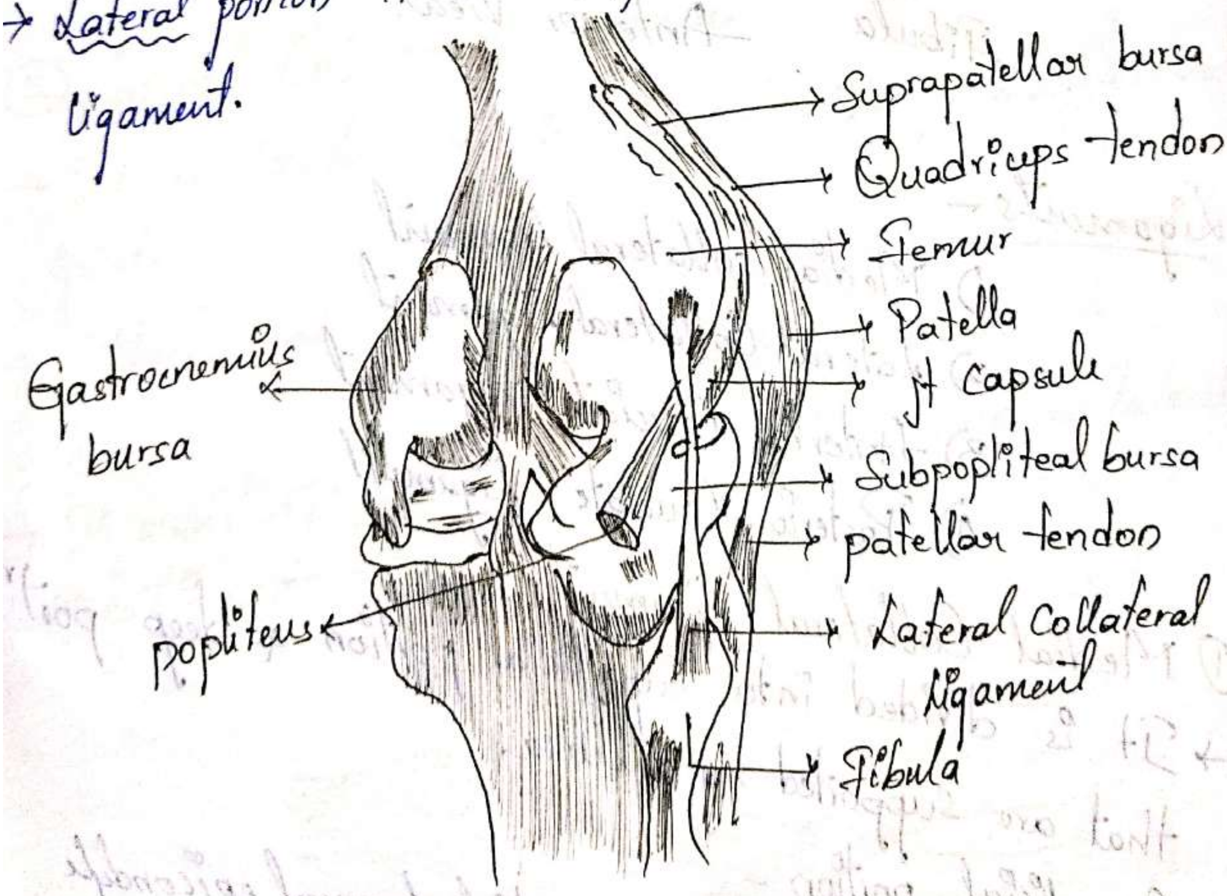
→ Joint capsule consist of 2 membrans

1) fibrous membrane

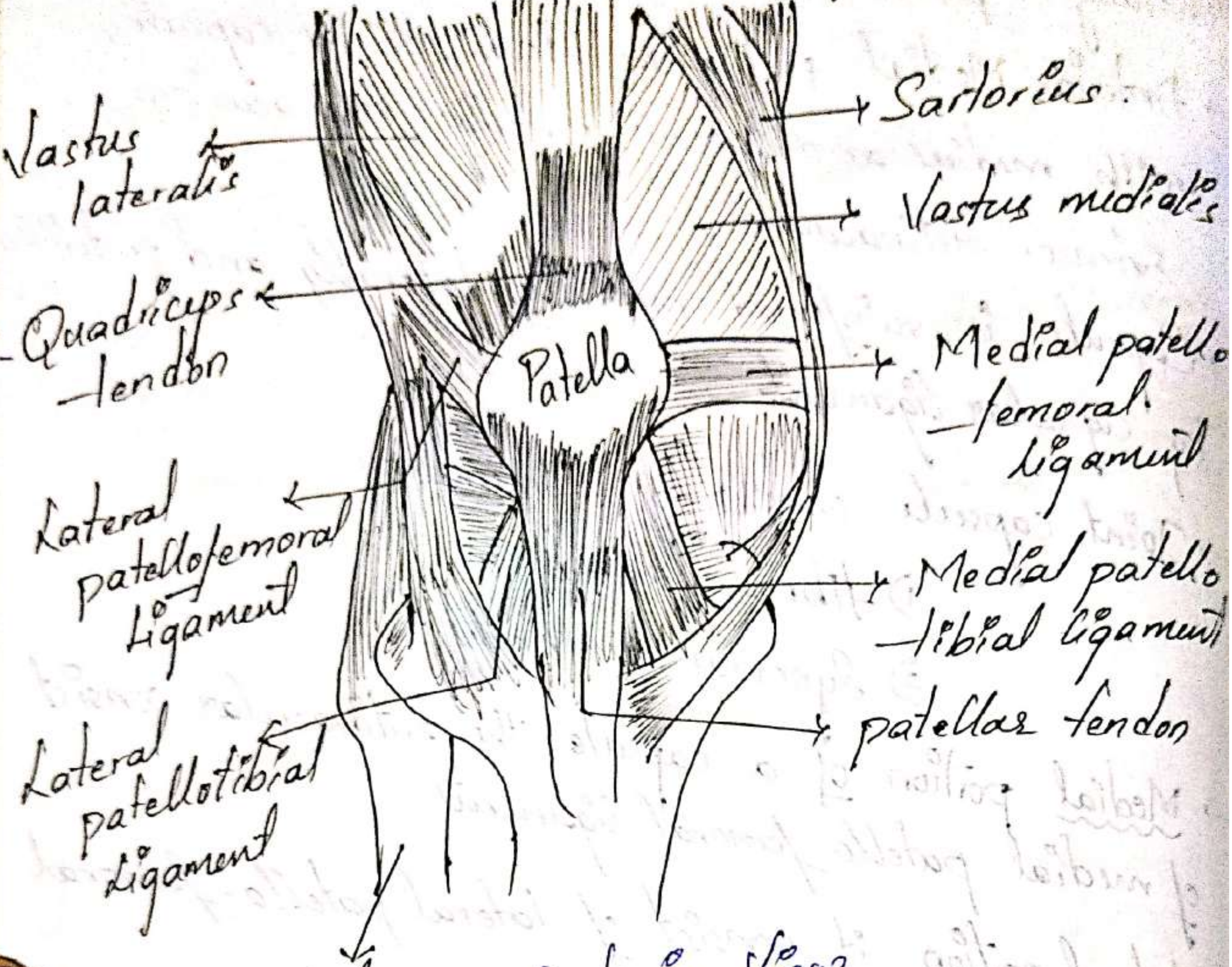
2) Synovial membrane.

→ Medial portion of a capsule the retinacular consist of medial patello-femoral ligament.

→ Lateral portion it consist of lateral patello-femoral ligament.



Posterior View



Anterior View.

Ligaments -

1) Medial collateral Ligament
2) Lateral Collateral Ligament
3) Anterior Cruciate Ligament
4) Posterior Cruciate Ligament

(1) Medial Collateral Ligament -

→ It is divided into Superficial portion & deep portion that are supported by a bursa.

→ Superficial portion -
proximally - from medial femoral epicondyle
distally - medial ascept of proximal tibia.

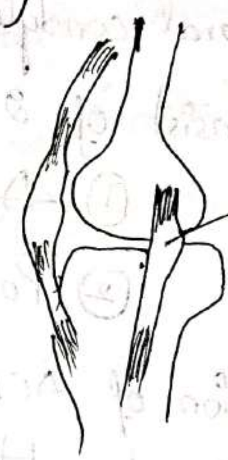
→ Deep portion -

Arises from inferior ascepts of the medial femoral condyle and inserts on the proximal of the medial tibial plateau.

→ The knee it is better able to resists <u>valgus stain</u> during extension than flexion due to the medial collateral ligament is taut in the knee extension

→ It has secondary rolling resisting anterior translation of tibia.

→ If the injury to the Medial Collateral ligament the load on Anterior cruciate ligament increases.

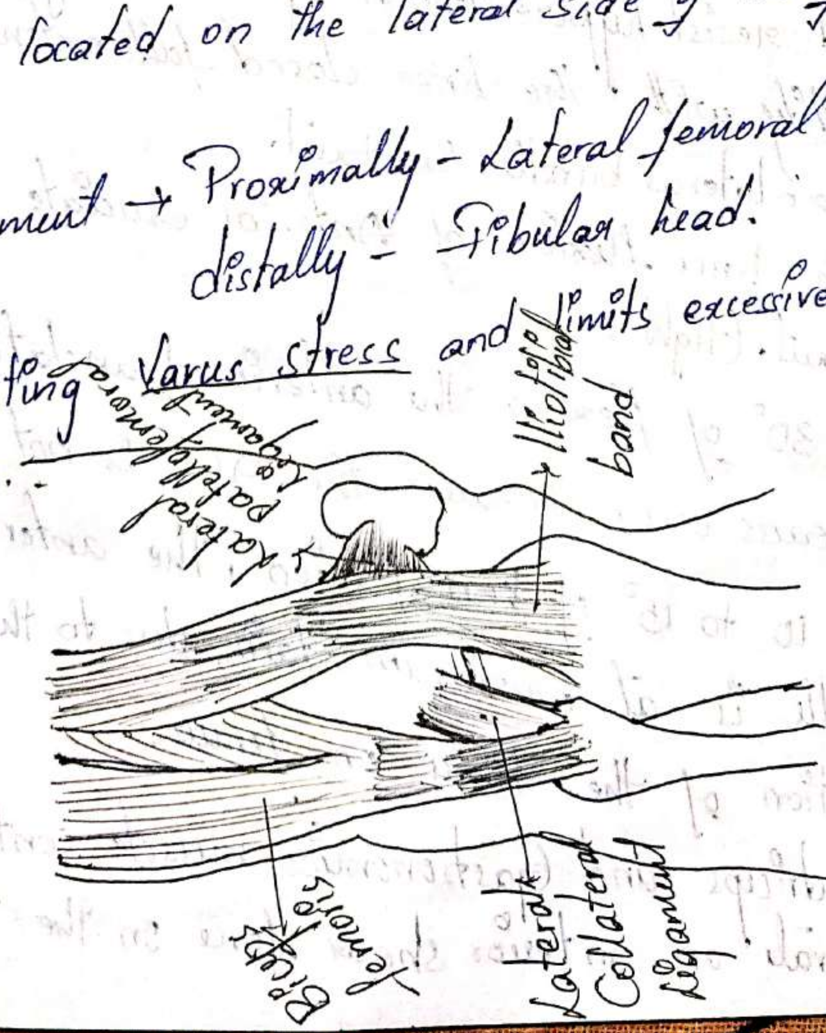


② Lateral Collateral Ligament -

→ It is located on the lateral side of tibiofemoral joint.

→ Attachement → Proximally - Lateral femoral condyle
distally - Fibular head.

→ It resisting <u>Varus stress</u> and limits excessive lateral rotation.



③ Anterior Cruciate Ligament -

→ It attached distally to the tibia on the lateral and anterior aspect of the medial intercondylar tibial spine.

→ It extends superiorly, laterally and posteriorly to attach to the posteromedial aspect of the lateral femoral condyle.

→ Consist of 2 bundles

① Anteriomedial bundle

② Posterolateral bundle

→ Function of ACL is the primary restraint against anterior translation of the tibia on the femur.

→ And resist hyperextension of knee & gives rotational stability with the knee closed full extension, the posterolateral bundle is taut.

→ With knee flexion of Anterior Cruciate ligament is taut. (tight)

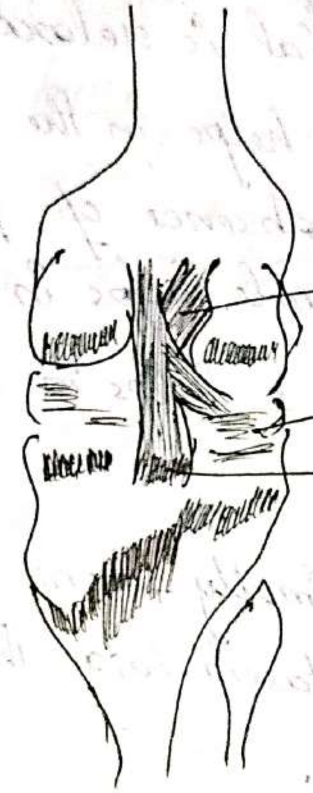
→ At 30° of flexion the anterior translation of tibia can ~~eatts~~ occur, because the ACL is not tensed.

→ At 10 to 15° of knee flexion, the anterior medial bundle is at maximum stain, due to the medial rotation of the tibia on femur.

→ Qudriceps and Gastronomics muscle contraction generate on anterior shear force on the tibia ps.

Stain on Anterior cruciate Ligament.

→ Hamstrings and Soleus contractⁿ creats posterior shear force on the tibia; hence relieve the stress on anterior cruciate ligament.



④ Posterior Cruciate Ligament –

→ Attaches distally the posterior tibial surface b/w posterior horn of 2 meniscus and travels superiorly some what anteriorly and attaches to the lateral ascepts of the medial condyles of the femur.

→ It is shorter and less oblique structure than ACL.

→ Cross Section area 120% to 150% greater than ACL

→ Consist of 2 bundels

① Posteromedial Bundle

② Antero~~medio~~ Lateral Bundle.

→ When knee is close to full extension, the larger and stronger anterolateral bundle is lax, posterior ~~lat~~ Medial bundle is taut. tight

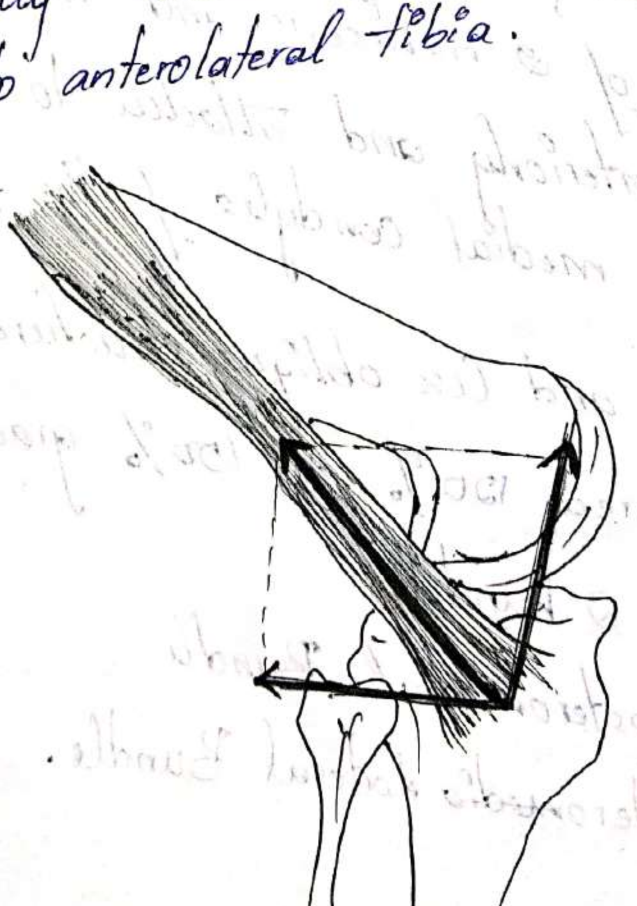
→ 80° to 90° of flexion, the Anterolateral is taut and posteromedial is relaxed.

→ Posteromedial bundle helps in the resisting translation of tibia absence of posterior cruciate ligament popliteal muscle helps in resisting posterior translation of tibia & helps in stability.

<u>Iliotibial Band</u> -

→ It is formed proximally from the fascia investing the tensor fascia lata, the gluteus maximus & gluteus medius muscle.

→ Attaches -
distally - lateral intermuscular septum &
inserted into anterolateral tibia.



Bursae -

→ 3 Bursae
① Suprapatellar bursa
② Subpopliteal bursa
③ Gastrocnemius bursa

→ Suprapatellar bursa -
Lies b/w quadriceps tendon & the anterior femur, superior to the patella.

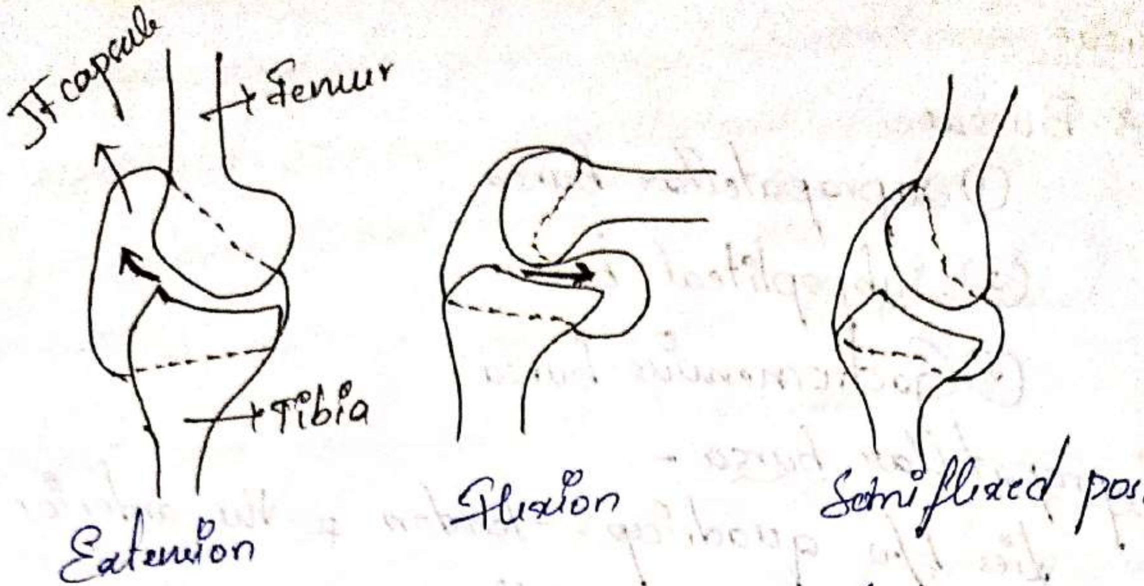
→ Subpopliteal bursa -
Lies b/w tendon of the popliteus muscle & lateral femoral condyle.

→ Gastrocnemius bursa -
Lies b/w tendon of the medial head of the gastrocnemius muscle and the medial femoral condyle.

→ Extension -
posterior capsule & ligament are taut, gastrocnemius and p. subpopliteal bursa are compressed.
Synovial fluid shifts anteriorly.

→ Flexion -
Suprapatellar bursa anteriorly is compressed & the fluid shift posteriorly.

→ Semiflexed position -
Synovial fluid is under the least amount of pressure.



→ There are 3 bursa inside jt capsule.

① Prepatellar bursa / Housemaid's knee.
② Infrapatellar bursa
③ Deep infrapatellar bursa.

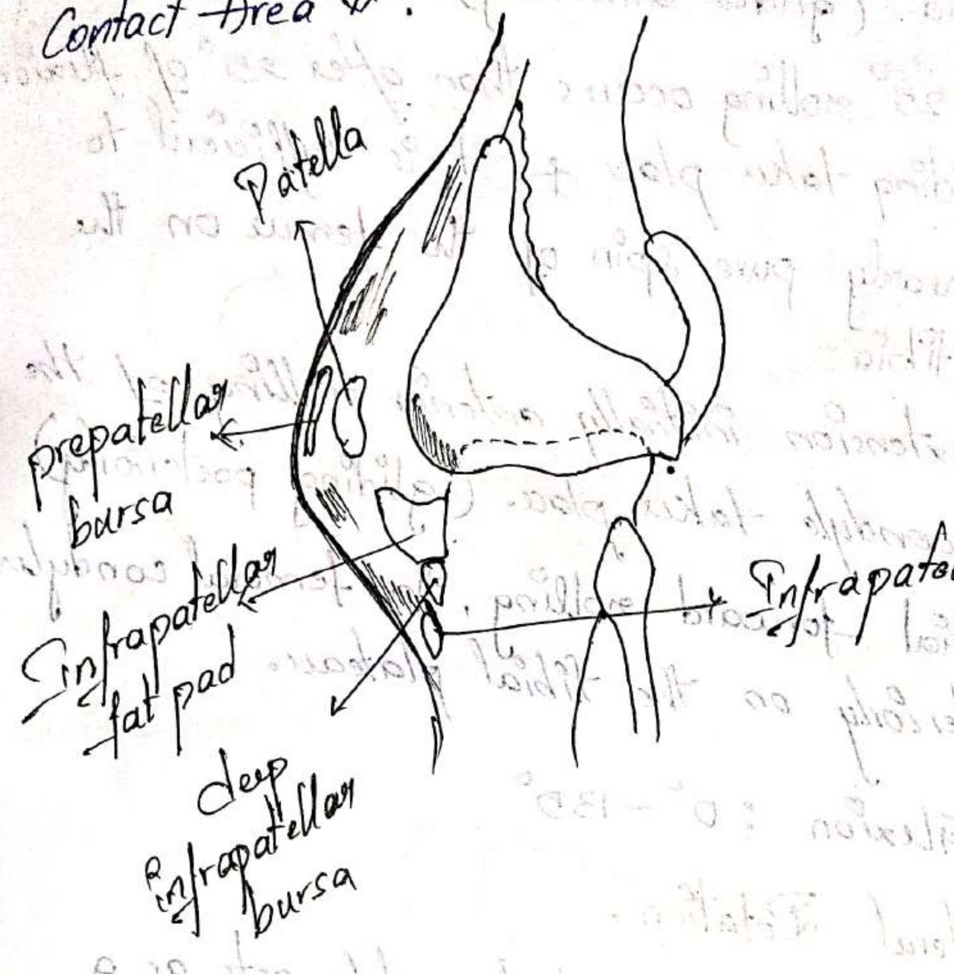
→ Prepatellar bursa -
Located b/w Skin & anterior surface of patella
free movement for flexion & Extension.

→ Infrapatellar bursa -
Lies inferior to patella, b/w patellar tendon
& overlying skin.

→ Deep infrapatellar bursa -
b/w patellar tendon & the tibial tuberosity.
They helps to reduce friction b/w the patellar tendon & the tibial tuberosity.

→ Bursa is seperated from the synovial cavity of the jt by the infrapatellar fat pad / Hoffa's fat pad.

→ Hoffa's fat pad is removed, patellofemoral Contact Area ↓s.



Joint Kinematics -

→ It has 3° of freedom.

Flexion / Extension

Medial / Lateral rotation

(Valgum) Abduction / Adduction (Varus)

Flexion + Extension -

~~Deties~~ - ~~no~~ → Hyperextension of knee - Genu Recurvatum

→ Femoral condyles roll + glide over the tibial plateaus.

→ The large articular surface of the femur + the relatively small tibial condyles create a potential problem as the femur begins to flex on the fixed tibia.

→ During flexion the femoral condyles rolls posteriorly on the tibia. (glides anteriorly).

→ Upto 0° - 25° rolling occurs than after 25° of flexion anterior gliding takes place & it is sufficient to create a nearly pure spin of the femur on the posterior tibia.

→ During extension initially anterior rolling of the femoral condyle takes place. (gliding posteriorly)

→ After initial forward rolling, the femoral condyles glide posteriorly on the tibial plateau.

ROM - flexion : 0° - 135°

Medial / Lateral Rotation -

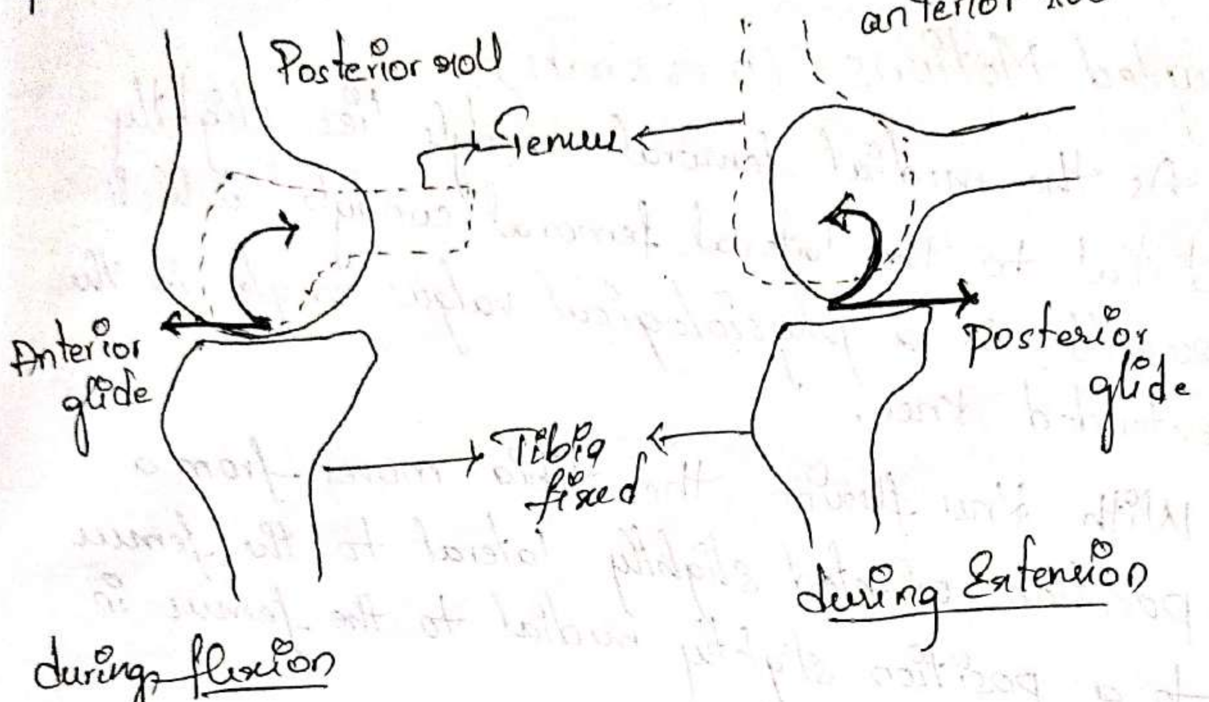
→ During rotations the medial condyle acts as a pivot point while the lateral condyles move through a greater arc of motion.

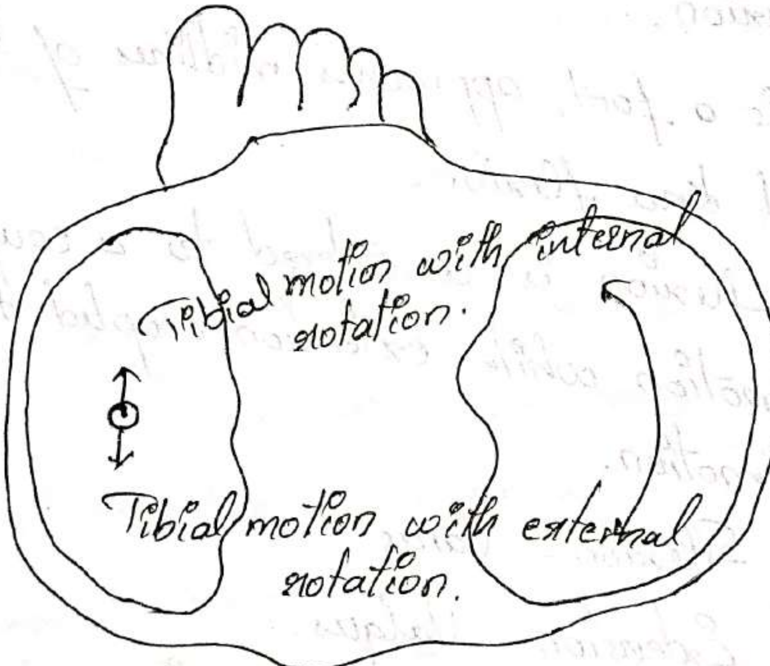
→ During lateral rotation the medial tibial condyle moves slightly on the fixed medial femoral condyle & the lateral tibial condyle move posteriorly at the large distance on the fixed lateral femoral condyle.

· ROM - Medial Rotation - 15°

Lateral Rotation - 20°

→ During medial rotation the medial condyle is fixed & tibia moves only slightly but the lateral tibia condyles moves larger distance anteriorly on fixed femoral condyle.





Medial & lateral Rotation.

Abduction and Adduction -

→ Abduction - Valgus
Adduction - Varus

→ Frontal plane ROM is typically only 8° at full extension, and 13° with 20° of knee flexion.

→ Excessive frontal plane movement indicates ligamentus insufficiency.

Coupled Motions – (5 or 2 mrks)

→ As the medial femoral condyle lies slightly distial to the lateral femoral condyle, which results in a physiological valgus angle in the extended knee.

→ With knee flexion, the tibia moves from a position oriented slightly lateral to the femur to a position slightly medial to the femur in full flexion.

→ That is a foot, approches midlines of the body with ~~H~~ knee flexion.

→ Hence, flexion is considered to a couple to a Varus motion while extension coupled to the Valgus motion.

Flexion – Varus

Extension – Valgus.

Locking and Unlocking mechanism of knee

(or)

Screw Home Mechanism.

→ There is a ~~tibi~~ Obligatory lateral rotion of the tibia that accompanies the final stages of knee extension that is not voluntary.

→ This coupled motion is reffered as Automatic or terminal rotation.

→ During, the last 30° of extension in non. wt-bearing knee extension, the lateral tibial plateau or femoral condyle pair completes its rolling & gliding motion

→ Extension continues, the medial plateau continues to roll & glide anteriorly after the lateral side of the plateau has halted.

→ Due to this continued medial condyle motion the lateral tibial condyle rotates laterally during extension & bringing the knee jt in close pack position.

→ The tibial spine large into the intercondylar notch & know the ligament are taught, there automatic rotiation is known as Locking or Screw home mechanism.

→ To initiate the knee flexion from complete extens^n the knee must be unlocked during flexion the medial and lateral tibial plateau rol & glide side in posterior direction.

→ The Smaller tibial plateau completes its roll & glide and in larger tibial plateau medial completes roll + glide posteriorly.

→ Making the lateral tibial plateau to rotate medially this is known as unlocking.

popliteus - Unlocking

Vastus medialis - Locking.

Muscles -

Knee Flexor Group -

Semimembranosus ✓

Semitendinosus ✓

Biceps femoris.

Sartorius ✓

Gracilis ✓

Popliteus

Gastronemius ✓

→ Two-jt muscles are Semimembranosus, Semitendinosus, Sartorius, Gracilis, Gastronemius.

→ Medially rotate the tibia on fixed femur are Semimembranosus, Semitendinosus, Sartorius, Gracilis, popliteus.

→ Lateral rotating the tibia ~~are~~ is biceps femoris

→ Valgus movement at knee are capable by biceps femoris, lateral head of gastronemius and popliteus.

→ Varus moments are Semimembranosus, Semi-tendinosus, medial head of the gastrocnemius.

Knee Extensor Group -

Rectus Femoris
Vastus Lateralis
Vastus Medialis
Vastus Intermedius

} Quadriceps muscles

→ Two-jt muscle is rectus femoris.

→ Purest knee extensor is Vastus intermedius.

→ There are 2 types of fibers in vastus medials.

(1) Vastus Medialis longus → pulling the patella medially.

(2) Vastus Medialis Oblique.

Quadriceps muscle function -

→ Strongly influenced by the patella.

→ Patella act as a anatomically pully, It defects the actions live of quadriceps.

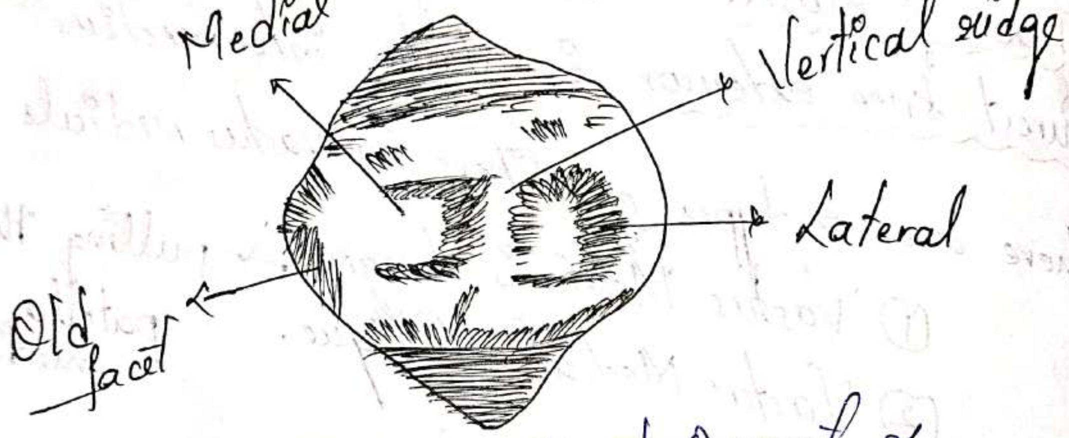
→ During early flexion, the patella acts in ring quadricep angle, However with complete flexion the angle deminishes.

→ As the patella fix in the intercondylar notch.

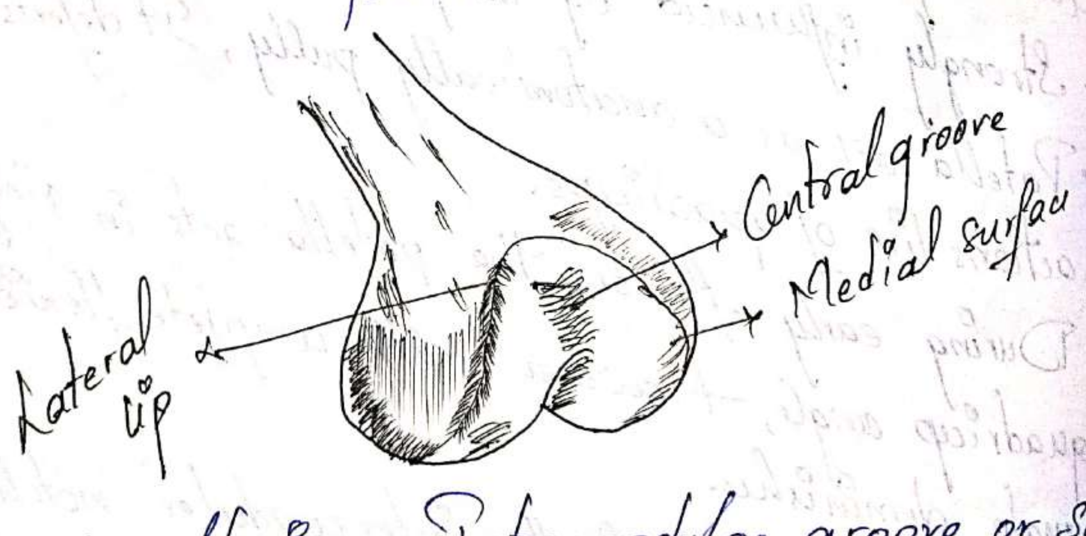
→ Quadriceps weakness or patellectomy leads to "Quad lag".

Patellofemoral Joint :-

→ Patella is the largest sesamoid bone.
→ Patella is inverted triangle with its apex directed inferiorly or downward
→ Posterior surface is divided by a vertical ridge.
→ Ridge present in center of patella.



Posterior ~~Ascpt~~ Aspect of patella



→ During flexion - Intercondylar groove or Sul
→ Patella is attached with tibial tuberosity wi patellar tendon.
→ Patellar tendon act as a pulley for quadricep muscle.

→ Quadricep-tendons reduces the friction b/w patella and femoral condyle.

Articular Surface -

→ Ratio of patella and patellar tendon is 1:1.

→ A marked high position of the patella produced due to long tendon of patella - Patella alta.

→ A marked short tendon produce abnormal low position of patella - Patella Baja.

→ Both this act as instability of patella.

→ During full flexion, the patella largest in the intercondylar groove & the contact is on the lateral & odd facet, the medial facet is out of contact.

Motions of the patella

→ It is a plane variety of jt.

→ Motions are : patellar flexion / Extension
Medial / Lateral shift
Medial / Lateral tilt
Medial / Lateral rotation

Patellar flexion -

→ Occurs in Sagittal plane.

→ Patella travels down the intercondylar groove of the femur is termed patellar flexion.

Patellar Extension -

→ Knee extension brings the patella back to its original position in the femoral sulcus.

→ This patellar motion of gliding superiorly while rotating up & around the femoral condyles is referred to as patellar extension.

Lateral / Medial Patellar ~~Shift~~ tilt :-

→ The patella tilts around a longitudial axis.

→ LPT occurs when the lateral edge of the patella approximates the surface of the lateral femoral condyle.

→ MPT occurs when the medial edge of patella moves toward the medial femoral condyle.

Lateral & Medial patellar Shift -

→ These are traslations (gliding) that occur in a plane of motion, rather than rotation around an axis.

→ LPS is defined as patella, moving toward the lateral femoral condyle in frontal plane.

Medial / Lateral rotation of patella -

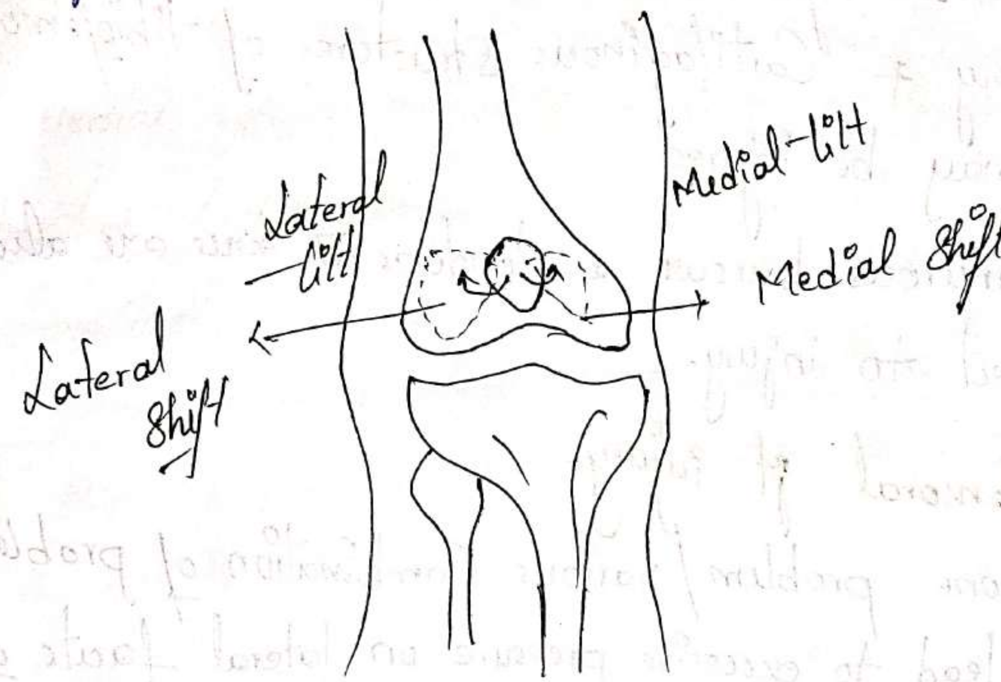
→ Anteroposterior axis.

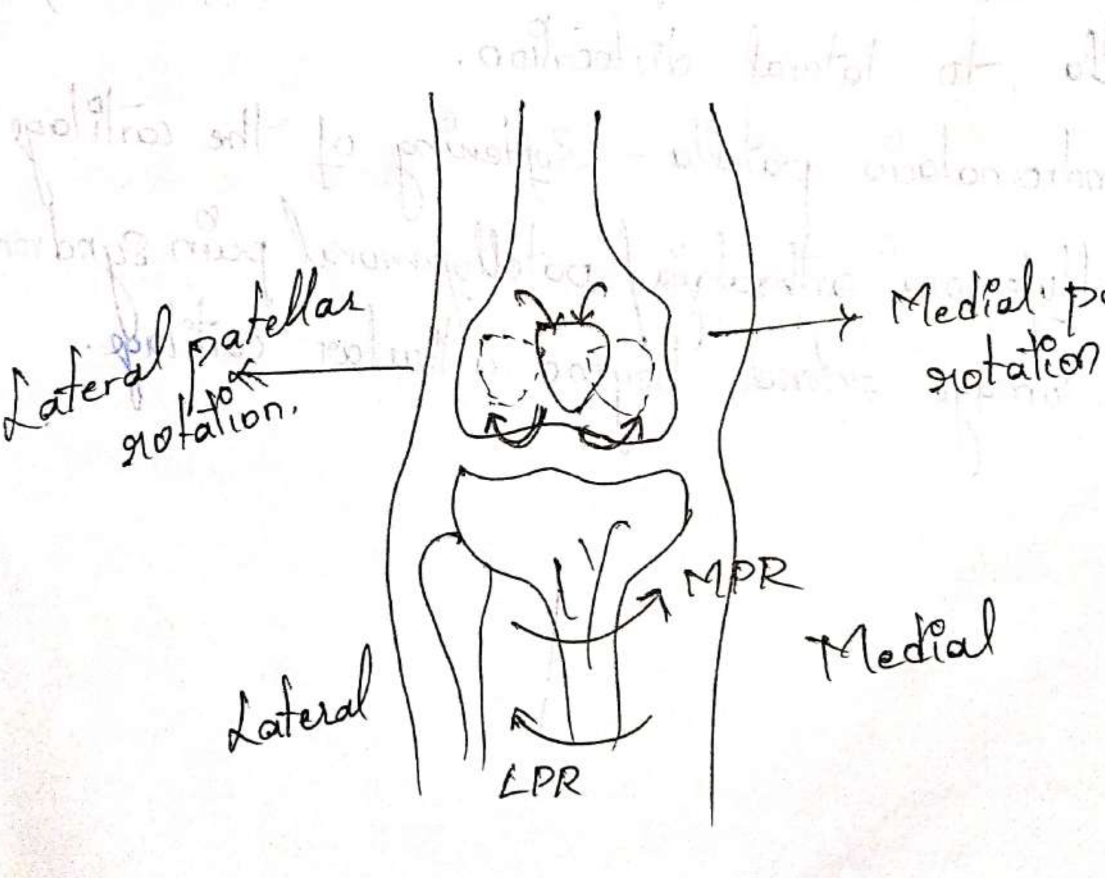
→ MR :- Patella spins around the perpendicular axis.

- Apex of patella pointing toward the medial femoral condyle.

- Base moving closer to the lateral femoral Condyle.

→ The patella rotates laterally to keep apex aligned with tibial tuberosity.





Injury and Disease

Tibiofemoral jt Injury-

→ Tremendous forces applied throughout the knee have to bear injuries & degenerative damage.

1) Meniscal injuries.

2) Bony & Cartilaginous structures of tibiofemoral jt may be injured.

3) Numerous bursar & tendons at knee are also subject to injury.

Patellofemoral jt injury-

→ Any one problem / various Combination of problems may lead to excessive pressure on lateral facets of patella, to lateral dislocation.

* Chondromalacia patella - Softening of the cartilage
* Patellofemoral arthralgia / patellofemoral pain syndrome : Damage extends beyond articular cartilage.